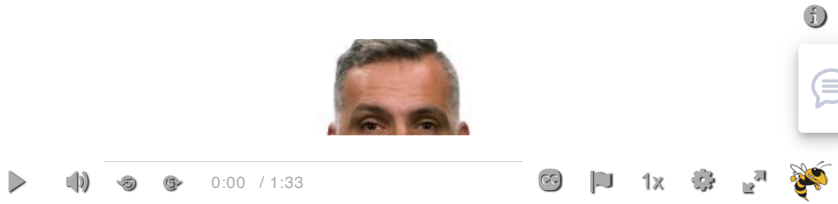


Lesson Thirteen Overview



Throughout the lecture, we are giving you some “**food for thought**” -- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will:

- Understand some major questions and methods in "network inference"
- Learn key network sampling strategies
- Review link prediction statistical methods
- Infer association networks based on correlation metrics
- Understand the principles of network tomography through theory and case studies

Required and Recommended Reading

Required Reading

- [Sampling and estimation in network graphs \(https://link.springer.com/chapter/10.1007/978-0-387-88146-1_5\)](https://link.springer.com/chapter/10.1007/978-0-387-88146-1_5), by Eric D.Kolaczyk, 2009
- [Network inference with confidence from multivariate time series \(https://journals.aps.org/pre/abstract/10.1103/PhysRevE.79.061916\)](https://journals.aps.org/pre/abstract/10.1103/PhysRevE.79.061916), by Mark Kramer et al, 2009
- [Network topology inference \(https://link.springer.com/chapter/10.1007%2F978-0-387-88146-1_7\)](https://link.springer.com/chapter/10.1007%2F978-0-387-88146-1_7), by Eric D.Kolaczyk, 2009

Recommended Reading

- [What is the real size of a sampled network? The case of the Internet \(https://journals.aps.org/pre/abstract/10.1103/PhysRevE.75.056111\)](https://journals.aps.org/pre/abstract/10.1103/PhysRevE.75.056111) by Fabian Viger et al., 2007

Assignments

📁 L13: Knowledge Check

L13: Quiz